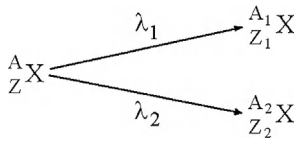
5.14. BRANCHING DECAY

Several times in this chapter the possibility of competing modes of decay has been noted; see e.g. Figure 1.2. In such competition, termed *branching decay* (see e.g. Fig. 5.5(g)), the parent nuclide may decay to two or more different daughter nuclides: e.g.



(5.71)

where for each branching decay a *partial decay constant* can be determined. These constants are related to the total observed decay constant for the parent nuclide as

$$\lambda_{\text{tot}} = \lambda_1 + \lambda_2 + \ldots \tag{5.72}$$

Each mode of decay in branching may be treated separately; the decay in an individual branch has a half-life based on the partial decay constant. Since only the total decay constant (the rate with which the mother nuclide, ${}^{A}_{Z}\text{X}$ in (5.71), decays) is observable directly, partial decay constants are obtained by multiplying the observed total decay constant by the fraction of parent decay corresponding to that branch. ${}^{64}\text{Cu}$ decays 43% by electron capture, 38% by negatron emission, and 19% by positron emission. The observed total decay constant is equal to $0.0541\ \text{h}^{-1}$ based on the half-life of 12.8 h. The partial constants are:

$$\lambda_{\text{EC}} = 0.43 \times 0.0541 = 0.0233\ \text{h}^{-1} \tag{5.73a}$$

$$\lambda_{\beta^-} = 0.38 \times 0.0541 = 0.0206\ \text{h}^{-1} \tag{5.73b}$$

$$\lambda_{\beta^+} = 0.19 \times 0.0541 = 0.0103\ \text{h}^{-1} \tag{5.73c}$$

These partial decay constants correspond to partial half-lives of 29.7 h for electron capture decay, 33.6 h for β^- decay, and 67.5 h for positron decay.

5.15. SUCCESSIVE RADIOACTIVE DECAY

There are many instances where a parent decays to a daughter which itself decays to a third species (i.e. a "grand-daughter"). The chains of radioactive decay in the naturally occurring heavy elements include as many as 10−12 successive steps (Fig. 1.2).

$$\text{X}_1 \xrightarrow{\lambda_1} \text{X}_2 \xrightarrow{\lambda_2} \text{X}_3 \xrightarrow{\lambda_3} \text{X}_4 \ldots \tag{5.74}$$

The net rate of formation of the daughter atoms X_2 is the difference between the rate of formation of the daughter and her rate of decay, i.e.

$$dN_2/dt = N_1\lambda_1 - N_2\lambda_2 \tag{5.75}$$

where N_1 and N_2 are the number of parent and of daughter atoms, and λ_1 and λ_2, the decay constants of the parent and daughter, respectively. The solution of this equation is

$$N_2 = [\lambda_1/(\lambda_2 - \lambda_1)]\, N_{1,0}\left(e^{-\lambda_1 t} - e^{-\lambda_2 t}\right) + N_{2,0}\, e^{-\lambda_2 t} \tag{5.76}$$

where $N_{1,0}$ and $N_{2,0}$ are the amounts of parent and daughter respectively at time $t = 0$. The first term in this equation tells us how the number of daughter nuclei vary with time as a consequence of the formation and subsequent decay of the daughter nuclei, while the second term accounts for the decay of those daughter nuclei that were present at $t = 0$. It should be noted that the successive radioactive decay is a special case of the more general question of irradiation yield and decay, cf Ch. 17.

Let us illustrate this relationship by an example among the naturally occurring radioactive decay series. In an old uranium mineral all the products in the decay chain can be detected (see Fig. 1.2). Suppose now that we use a chemical separation to isolate two samples, one containing only uranium and one containing only thorium (relation (1.3) and Fig. 1.1). At the time of separation, which we designate as $t = 0$, there are $N_{1,0}$ atoms of ^{238}U and $N_{2,0}$ atoms of ^{234}Th. In the thorium fraction, which is free from uranium, $N_{1,0} = 0$ and, therefore, the thorium atoms decay according to the last term in (5.76). This sample gives a simple exponential decay curve with a half-life of 24.1 d, as shown by the precipitate curve in Figure 1.1. The uranium fraction at $t = 0$ is completely free of thorium; i.e. $N_{2,0} = 0$. However, after some time it is possible to detect the presence of ^{234}Th. The change in the number of ^{234}Th with time follows the first term of (5.78); in fact, in Figure 1.1 the measurements detect only the ^{234}Th nuclide (β-emitting), since the detection system used is not sensitive to the α's from ^{238}U ($\psi_\alpha = 0$). The time of observation is much smaller than the half-life of the ^{238}U decay, so there will be no observable change in the number of atoms of uranium during the time of observation, i.e. $N_1 = N_{1,0}$. Further, since $t_{½}$ for $^{238}U \gg t_{½}$ for ^{234}Th, i.e. $\lambda_1 \ll \lambda_2$, we can simplify (5.76) to

$$N_2 = (\lambda_1/\lambda_2)N_{1,0}\left(1 - e^{-\lambda_2 t}\right) \tag{5.77}$$

According to this equation, the number of ^{234}Th atoms, N_2, increases with time with the half-life of ^{234}Th. In other words, after a period of 24.1 d there is 50% of the maximum value of ^{234}Th, after 48.2 d there is 75% of the final maximum value, etc. This is illustrated by the change in the uranium fraction activity in Figure 1.1. Further, from (5.77) we can see that the maximum value of thorium ($t = \infty$) is given by

$$N_2\,\lambda_2 = N_{1,0}\,\lambda_1 \tag{5.78}$$

These equations, based on $\lambda_1 \ll \lambda_2$, show that the amount of daughter atoms becomes constant after some time. At that time the rate of decay of the daughter becomes equal to the rate of decay of the parent, i.e. $A_2 = A_1$, but the amounts of the parent and the daughter are not equal since N_2 is much smaller than N_1. This condition of $A_2 = A_1$ is known as *secular equilibrium*, which is a misnomer since this is a steady state and not a true equilibrium situation. It is also common to speak of *radioactive equilibrium* in referring to this steady state condition.

We can calculate that at secular equilibrium for each gram of ^{238}U there will be present 1.44×10^{-11} g of ^{234}Th and 4.9×10^{-16} g of ^{234}Pa. Since the specific radioactivity of ^{238}U is 746 000 dpm/g, the decay rate of 4.9×10^{-16} g ^{234}Pa is also 746 000 dpm.

When the time of observation is very short compared to the half-life of the parent nuclide, as in secular equilibrium, no change in the decay rate of the parent is observed for many daughter half-lives. Our example of ^{137}Cs, which decays via the isomeric state ^{137m}Ba to ^{137}Ba, presents another case of "secular equilibrium". If we have an "old" sample in which radioactive equilibrium has been reached (older than ~15 min, since the $t_{½}$ of the daughter is 2.6 min), and separate the cesium from the barium by precipitation and filtration of $BaSO_4$, the activity measured from the precipitate will follow curve (1) in Figure 5.11. In the filtrate solution the activity from ^{137}Cs, curve (2), is unchanged

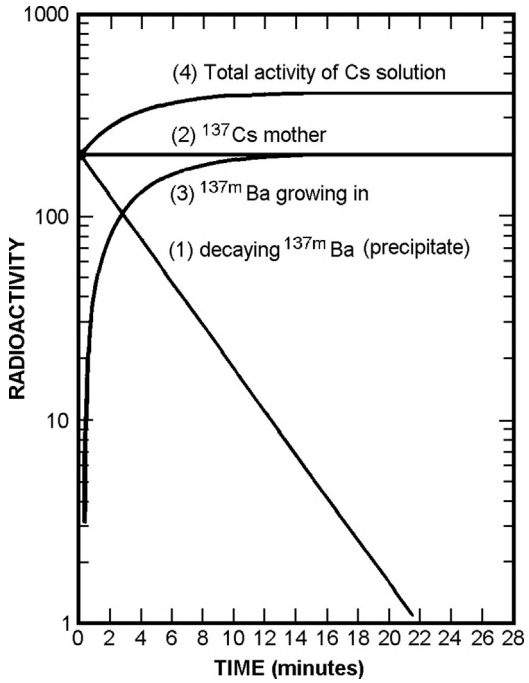


Figure 5.11 Case of radioactive equilibrium: successive decay chain ^{137}Cs($t_{½}$ 30 y) → ^{137m}Ba($t_{½}$ 2.6 min) → stable.

during our observation time. However, ^{137m}Ba grows into the solution, curve (3), so that the total activity of the solution, curves (2) plus (3), increases according to curve (4).

In many radioactive decay chains the half-life of the parent is longer than that of the daughter but it is short enough that a change in the disintegration rate of the parent is observable during the period of observation of the experiment. In such cases the system reaches the condition termed *transient equilibrium*. The length of time of observation of the activity of the sample may be the determining factor as to whether it appears to be transient or secular equilibrium. If a parent has a one month half-life and the observation of the change in decay rates of parent and daughter extends over an hour or even a few days, the data would follow the equation for secular equilibrium since the degree of change in the parent decay would be negligible. However, if the observation extends over a period of several weeks or months, then the change in the decay rate of the parent is significant and it would appear as transient equilibrium.

The case of transient equilibrium will be illustrated by an example, such as the decay chain

$$^{140}\mathrm{Ba}\left(\beta^{-}, 12.75\ \mathrm{d}\right)\ ^{140}\mathrm{La}\left(\beta^{-},\ 1.678\ \mathrm{d}\right)\ ^{140}\mathrm{Ce}\left(\mathrm{stable}\right) \tag{5.79}$$

^{140}Ba is one of the most important fission products. If we isolate barium, lanthanum grows into the sample. Figure 5.12 shows the decay of ^{140}Ba in curve (1), which follows

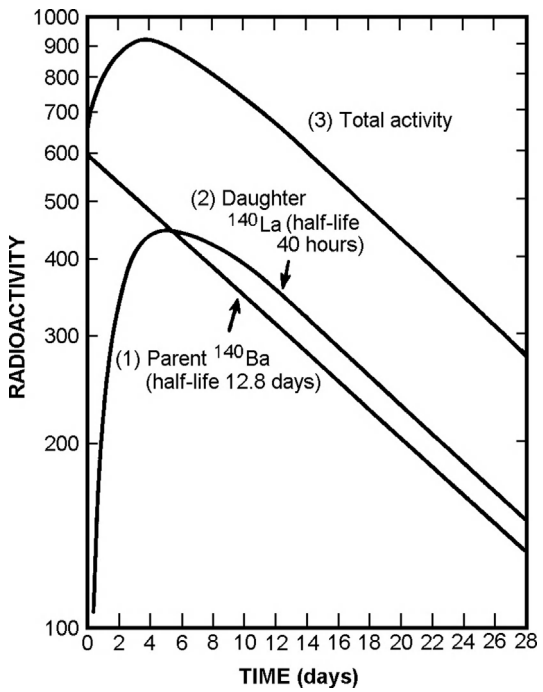


Figure 5.12 Case of transient equilibrium: successive decay chain ^{140}Ba($t_{½}$ 12.75 d) → ^{140}La($t_{½}$ 1.678 d) → stable.

the simple decay of (5.59). Curve (2) shows the activity of the daughter, for which the left half of eq. (5.76) is valid. Replacing decay constants by half-lives we can rewrite this equation as

$$A_2 = t_{½,1}/\left(t_{½,1} - t_{½,2}\right)A_{1,0}\left(\mathrm{e}^{-0.693t/t_{½,1}} - \mathrm{e}^{-0.693t/t_{½,2}}\right) \tag{5.80}$$

At $t \ll t_{½,1}$ ($t \ll 12.8$ d) the first exponential term is very close to 1 and A_2 increases proportional to $(1 - \mathrm{e}^{-0.693t/t_{½,2}})$; this is the increasing part of curve (2). At $t \ll t_{½,2}$ ($t \ll$ 40 h), the second exponential term becomes much smaller than the first one, and A_2 decreases proportional to $\mathrm{e}^{-0.693t/t_{½,1}}$. For this part of the curve we may write

$$N_2 = N_1\lambda_1/(\lambda_2 - \lambda_1) \tag{5.81}$$

which is the relation valid for transient equilibrium. The total activity of the barium sample, curve (3), is the sum of curves (1) and (2).

If the parent is shorter-lived than the daughter, the daughter activity grows to some maximum value and then decays with its own characteristic half-life. This contrasts to the case of transient equilibrium where the daughter has an apparent decay given by the half-life of the parent. An example of this is shown in Figure 5.13 for the decay chain

$$^{218}\mathrm{Po}\left(\alpha, 3\ \mathrm{min}\right)\ ^{214}\mathrm{Pb}\left(\beta^-, 27\ \mathrm{min}\right)\ ^{214}\mathrm{Bi} \tag{5.82}$$

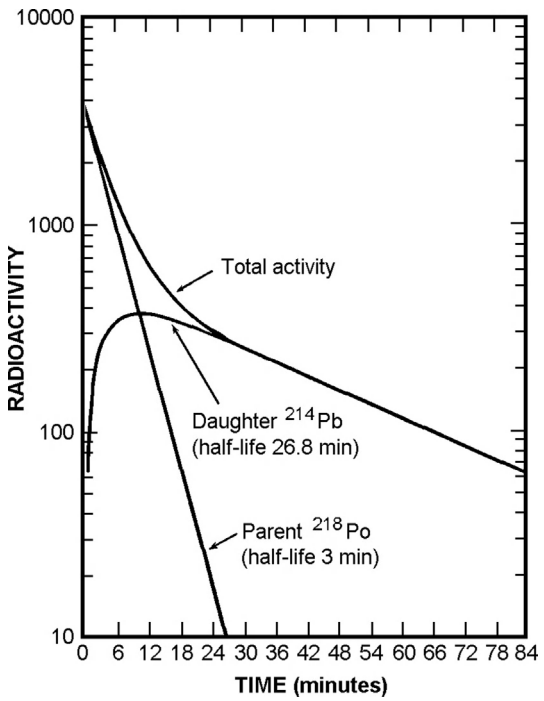


Figure 5.13 Case of no equilibrium: successive decay chain ^{218}Po($t_{½}$ 3 min) → ^{214}Pb($t_{½}$ 26.8 min) → stable.

The time necessary for obtaining the maximum daughter intensity in the non-equilibrium case of the shorter-lived parent is given by

$$t_{max} = (\lambda_2 - \lambda_1)^{-1} \ln(\lambda_2/\lambda_1) \tag{5.83}$$

5.16. RADIOISOTOPE GENERATORS

The growth of radioactive daughters frequently has practical significance. For example, in radiation therapy and diagnostic medicine it is preferable to use short-lived nuclides. In fact, it is preferable to conduct tracer experiments with short-lived nuclides as this eliminates the problem of disposal of residual radioactive waste after completion of the experiment. It is convenient to have a long-lived mother in storage from which a short-lived daughter can be removed as required for use in tracer work. A few examples of uses of such mother-daughter pairs are discussed; others are included in Table 5.1.

Such systems are called *radioisotope generators*. ^{222}Rn is sometimes used for the radiotherapeutic treatment of cancer. This product is isolated by separating it as a gas from the parent substance ^{226}Ra which is normally in the form of solid or a solution of $RaBr_2$.

Table 5.1 Some common radioactive milking pairs. The decay properties include decay energy (MeV), mode of decay and half-life

Mother nuclide	Decay properties	Daughter nuclide	Decay properties	Application
^{44}Ti	EC, γ; 47.3 y	^{44}Sc	1.5 β^+; 1.16 γ; 3.93 h	Teaching
^{68}Ge	EC; 270.8 d	^{68}Ga	1.9 β^+; 1.08 γ; 1.135 h	Medical
^{87}Y	EC; 3.35 d	^{87m}Sr	0.39 γ; 2.80 h	Medical & teaching
^{90}Sr	0.5 β^-; 28.5 y	^{90}Y	2.3 β^-; 2.671 d	Heat source[†], Calibration source
^{99}Mo	β^-, γ; 65.9 h	^{99m}Tc	0.14 γ; 6.0 h	Medical
^{113}Sn	EC, γ; 115.1 d	^{113m}In	0.39 γ; 1.658 h	Medical
^{132}Te	β^-, γ; 78.2 h	^{132}I	2.1 β^-, γ; 2.28 h	Medical
^{137}Cs	β^-, γ; 30.0 y	^{137m}Ba	0.66 γ; 2.55 m	Gamma radiography, Radiation sterilization[†]
^{140}Ba	β^-, γ; 12.75 d	^{140}La	β^-, γ; 1.678 d	Lanthanum tracer
^{144}Ce	β^-, γ; 284.9 d	^{144}Pr	3.0 β^-; 17.28 m	Calibration source
^{210}Pb	β^-, γ; 22.3 y	^{210}Bi	1.2 β^-; 5.01 d	Calibration source
^{226}Ra	α; 1600 y	^{222}Rn	α; 3.825 d	Medical
^{238}U	α; 4.468×10^9 y	^{234}Th	β^-, γ; 24.1 d	Thorium tracer

[†]Main use of mother substance in large amounts.